Exam Date & Time: 02-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

MAKEUP EXAMINATION-2025 JANUARY

Introduction to Biomedical Nanotechnology [BME 4053]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.

1A)	"After nucleation, Ostwald ripening can lead to nanoparticles with uniform size." Validate this statement by elucidating the process of Ostwald ripening and its impact on nanoparticle size distribution.	(2)
1B)	Differentiate between homogeneous and heterogeneous nucleation in the synthesis of nanoparticles. Illustrate each process with examples	(3)
1C)	A student is preparing ZnO nanoparticles using Zinc nitrate and NaOH as reactants. Explain multiple steps in the growth process of ZnO nuclei? Recommend a method to limit the growth of ZnO nanoparticles to form monosized ZnO nanoparticles.	(5)
2A)	Compare solgel and precipitation methods for nanomaterial synthesis.	(2)
2B)	Propose the two microemulsion method for the preparation of zinc oxide nanoparticles. Zinc nitrate and sodium hydroxide will react to form zinc oxide.	(3)
2C)	Develop a synthesis method for creating a porous 3D nanofiber mesh suitable for water filtration. Propose adjustments to experimental parameters that would allow control over the formation of nano- and microporous structures.	(5)
3A)	Explain the principle behind and operation modes of Scanning Tunneling microscopy (STM).	(2)
3B)	Analyze the differences in nanoparticle size measurements obtained from DLS and SEM, discussing factors that influence the results in each method.	(3)
3C)	Suggest the method to find out the crystallite size of a nanomaterial and explain the equation for the same.	(5)
4A)	Explain the procedure to obtain the absorption spectra of zinc sulphide doped with manganese quantum dots.	(2)
4B)	Describe any three Physical Vapor Deposition (PVD) methods for nano-synthesis.	(3)
4C)	Propose four design strategies for developing Fluorescence Resonance Energy Transfer	(5)

(FRET)-based optical biosensors, discussing how each approach influences sensor performance and specificity.

5A) Suggest the advantages of Superparamagnetic iron oxide nanoparticles (SPIONs) over iron oxide for heat induced drug delivery (2)

5B) Design a nanoparticle system for photodynamic therapy and deliver a chemotherapeutic drug to the tumor tissues. Justify the design and working principle. (3)

5C) Design a nanoparticle contrast agent (a single nanosystem) for CT and SPECT multimodal imaging. Justify the design. (5)

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